BOUNDING THE NUMBER OF SUBOBJECTS OF AN OBJECT MODULO A GENERATOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all three assertions, the encoding used in the last one being defined explicitly in the span.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Generators and the maps out of them. Same chapter, the definition of a generator and the discussion of the maps from the generator to an object. Supplies the set of morphisms from the generator that the proof counts and uses to encode subobjects.
- Subobjects and their inclusions. Same chapter, the description of subobjects and of the factorisation of a morphism through a subobject. Supplies the notion of subobject and the statement that agreement of the sets of morphisms factoring through two subobjects forces the subobjects to be equal.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category with a generator U and X an object of $\mathcal{A}$. If κ is the cardinality of $\mathrm{Mor}(U, X)$ then*

1. *There does not exist a strictly increasing (or strictly decreasing) chain of subobjects of X indexed by a cardinal bigger than κ.*
2. *If α is an ordinal of cofinality $> \kappa$ then any increasing (or decreasing) sequence of subobjects of X indexed by α is eventually constant.*
3. *The cardinality of the set of subobjects of X is $\leq 2^\kappa$.*

Proof. For (1) assume $\kappa' > \kappa$ is a cardinal and assume X_i, $i \in \kappa'$ is strictly increasing. Then take for each i a $\phi_i \in \mathrm{Mor}(U, X)$ such that ϕ_i factors through X_{i+1}

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but not through X_i. Then the morphisms ϕ_i are distinct, which contradicts the definition of κ.

Part (2) follows from the definition of cofinality and (1).

Proof of (3). For any subobject $Y \subset X$ define $S_Y \in \mathcal{P}(\mathrm{Mor}(U, X))$ (power set) as $S_Y = \{\phi \in \mathrm{Mor}(U, X) : \phi)$ factors through $Y\}$. Then $Y = Y'$ if and only if $S_Y = S_{Y'}$. Hence the cardinality of the set of subobjects is at most the cardinality of this power set. $\square$